

Log File Analyzer for Intrusion Detection.

What you'll build

- parses Apache access logs and SSH auth logs,
- detects SSH brute-force, web scanning, and possible DoS,
- visualizes traffic (top IPs, requests per minute),
- cross-references a blacklist,
- exports alerts to CSV/JSON.

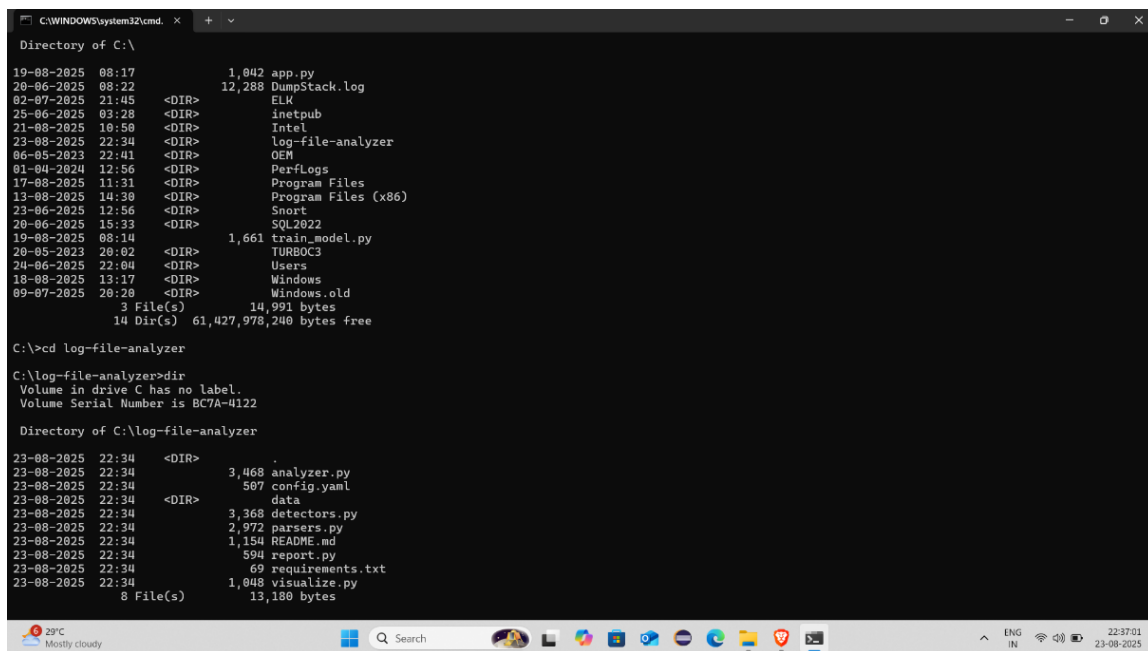
Status of My Project (Current Phase):

Environment Setup Completed:

Created a Python virtual environment, installed dependencies (pandas, matplotlib, PyYAML, etc.).

Set up the environment

- Create a venv and install deps:



```
C:\WINDOWS\system32\cmd. x + v
Directory of C:\
19-08-2025 08:17      1,042 app.py
20-06-2025 08:22    12,288 DumpStack.log
02-07-2025 21:45    <DIR>      ELK
25-06-2025 03:28    <DIR>      inetpub
21-08-2025 10:50    <DIR>      Intel
23-08-2025 22:34    <DIR>      log-file-analyzer
06-05-2023 22:41    <DIR>      OEM
01-04-2024 12:56    <DIR>      Perflogs
17-08-2025 11:31    <DIR>      Program Files
13-08-2025 14:30    <DIR>      Program Files (x86)
23-06-2025 12:56    <DIR>      Snort
20-06-2025 15:33    <DIR>      SQL2022
19-08-2025 08:14      1,661 train_model.py
20-05-2023 20:02    <DIR>      TURBOC3
20-06-2025 22:04    <DIR>      Users
18-08-2025 13:17    <DIR>      Windows
09-07-2025 20:20    <DIR>      Windows.old
          3 File(s)      14,991 bytes
          14 Dir(s)  61,427,978,240 bytes free

C:\>cd log-file-analyzer

C:\log-file-analyzer>dir
Volume in drive C has no label.
Volume Serial Number is BC7A-4122

Directory of C:\log-file-analyzer

23-08-2025 22:34    <DIR>
23-08-2025 22:34      3,468 analyzer.py
23-08-2025 22:34      597 config.yaml
23-08-2025 22:34    <DIR>      data
23-08-2025 22:34      3,368 detectors.py
23-08-2025 22:34      2,972 parsers.py
23-08-2025 22:34      1,154 README.md
23-08-2025 22:34      598 report.py
23-08-2025 22:34       69 requirements.txt
23-08-2025 22:34      1,048 visualize.py
          8 File(s)      13,180 bytes
```

```
C:\WINDOWS\system32\cmd. x + v
C:\log-file-analyzer>pip install -r requirements.txt
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas>=2.0.0 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-pa
ckages\python312\site-packages (from -r requirements.txt (line 1)) (2.2.1)
Collecting matplotlib>=3.7.0 (from -r requirements.txt (line 2))
  Downloading matplotlib-3.10.5-cp312-cp312-win_amd64.whl.metadata (11 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache
\local-packages\python312\site-packages (from -r requirements.txt (line 3)) (2.8.2)
Collecting PyYAML>=6.0.1 (from -r requirements.txt (line 4))
  Downloading PyYAML-6.0.2-cp312-cp312-win_amd64.whl.metadata (2.1 kB)
Requirement already satisfied: numpy<2,>=1.26.0 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local
-packages\python312\site-packages (from pandas>=2.0.0->-r requirements.txt (line 1)) (1.26.4)
Requirement already satisfied: pytz>=2020.1 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-pac
kages\python312\site-packages (from pandas>=2.0.0->-r requirements.txt (line 1)) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-p
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Collecting contourpy>=1.0.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading contourpy-1.3.3-cp312-cp312-win_amd64.whl.metadata (5.5 kB)
Collecting cycler>=0.10 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading fonttools-4.59.1-cp312-cp312-win_amd64.whl.metadata (111 kB)
    111.1/111.1 kB 642.5 kB/s eta 0:00:00
Collecting kiwisolver>=1.3.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading kiwisolver-1.4.9-cp312-cp312-win_amd64.whl.metadata (6.4 kB)
Collecting packaging>=20.0 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pillow>=8 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading pillow-11.3.0-cp312-cp312-win_amd64.whl.metadata (9.2 kB)
Collecting pyparsing>=2.3.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading pyparsing-3.2.3-py3-none-any.whl.metadata (5.0 kB)
Requirement already satisfied: six>=1.5 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-package
s\python312\site-packages (from python-dateutil>=2.8.2->-r requirements.txt (line 3)) (1.16.0)
Download matplotlib-3.10.5-cp312-cp312-win_amd64.whl (8.1 MB)
    8.1/8.1 MB 320.1 kB/s eta 0:00:00
Download PyYAML-6.0.2-cp312-cp312-win_amd64.whl (156 kB)
    156.3/156.3 kB 781.3 kB/s eta 0:00:00
Download contourpy-1.3.3-cp312-cp312-win_amd64.whl (226 kB)
    226.6/226.6 kB 816.3 kB/s eta 0:00:00
Download cycler-0.12.1-py3-none-any.whl (8.3 kB)
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C:\WINDOWS\system32\cmd. x + v
-ackages\python312\site-packages (from pandas>=2.0.0->-r requirements.txt (line 1)) (1.26.4)
Requirement already satisfied: pytz>=2020.1 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-pac
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Requirement already satisfied: tzdata>=2022.7 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-p
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  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading fonttools-4.59.1-cp312-cp312-win_amd64.whl.metadata (111 kB)
    111.1/111.1 kB 642.5 kB/s eta 0:00:00
Collecting kiwisolver>=1.3.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading kiwisolver-1.4.9-cp312-cp312-win_amd64.whl.metadata (6.4 kB)
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  Downloading packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pillow>=8 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading pillow-11.3.0-cp312-cp312-win_amd64.whl.metadata (9.2 kB)
Collecting pyparsing>=2.3.1 (from matplotlib>=3.7.0->-r requirements.txt (line 2))
  Downloading pyparsing-3.2.3-py3-none-any.whl.metadata (5.0 kB)
Requirement already satisfied: six>=1.5 in c:\users\hari6\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-package
s\python312\site-packages (from python-dateutil>=2.8.2->-r requirements.txt (line 3)) (1.16.0)
Download matplotlib-3.10.5-cp312-cp312-win_amd64.whl (8.1 MB)
    8.1/8.1 MB 320.1 kB/s eta 0:00:00
Download PyYAML-6.0.2-cp312-cp312-win_amd64.whl (156 kB)
    156.3/156.3 kB 781.3 kB/s eta 0:00:00
Download contourpy-1.3.3-cp312-cp312-win_amd64.whl (226 kB)
    226.6/226.6 kB 816.3 kB/s eta 0:00:00
Download cycler-0.12.1-py3-none-any.whl (8.3 kB)
    2.3/2.3 MB 885.2 kB/s eta 0:00:00
Download fonttools-4.59.1-cp312-cp312-win_amd64.whl (2.3 MB)
    73.9/73.9 kB 451.5 kB/s eta 0:00:00
Download kiwisolver-1.4.9-cp312-cp312-win_amd64.whl (73 kB)
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Download packaging-25.0-py3-none-any.whl (66 kB)
    7.0/7.0 MB 418.8 kB/s eta 0:00:00
Download pillow-11.3.0-cp312-cp312-win_amd64.whl (7.0 MB)
    111.1/111.1 kB 431.1 kB/s eta 0:00:00
Installing collected packages: PyYAML, pyparsing, pillow, packaging, kiwisolver, fonttools, cycler, contourpy, matplotlib
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C:\WINDOWS\system32\cmd. x + v
Downloading numpy-2.3.2-cp312-cp312-win_amd64.whl.metadata (60 kB)
Collecting pytz>=2020.1 (from pandas>=2.0.0->r requirements.txt (line 1))
Downloading pytz-2025.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas>=2.0.0->r requirements.txt (line 1))
Downloading tzdata-2025.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting contourpy>=1.0.1 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached contourpy-1.3.3-cp312-cp312-win_amd64.whl.metadata (5.5 kB)
Collecting cycycler>=0.10 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached cycycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.22.0 (from matplotlib>=3.7.0->r requirements.txt (line 2))
Using cached fonttools-4.59.1-cp312-cp312-win_amd64.whl.metadata (111 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib>=3.7.0->r requirements.txt (line 2))
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Collecting six>=1.5 (from python-dateutil>=2.8.2->r requirements.txt (line 3))
Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Downloading pandas-2.3.2-cp312-cp312-win_amd64.whl (11.0 MB)
11.0/11.0 MB 7.0 kB/s eta 0:00:00
Using cached matplotlib-3.10.5-cp312-cp312-win_amd64.whl (8.1 MB)
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Using cached PyYAML-6.0.2-cp312-cp312-win_amd64.whl (156 kB)
Using cached contourpy-1.3.3-cp312-cp312-win_amd64.whl (226 kB)
Using cached cycycler-0.12.1-py3-none-any.whl (8.3 kB)
Using cached fonttools-4.59.1-cp312-cp312-win_amd64.whl (2.3 MB)
Using cached kiwisolver-1.4.9-cp312-cp312-win_amd64.whl (73 kB)
Downloading numpy-2.3.2-cp312-cp312-win_amd64.whl (12.8 MB)
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Using cached packaging-25.0-py3-none-any.whl (66 kB)
Using cached pillow-11.3.0-cp312-cp312-win_amd64.whl (7.0 MB)
Using cached pyparsing-3.2.3-py3-none-any.whl (111 kB)
Downloading pytz-2025.2-py2.py3-none-any.whl (509 kB)
Downloading tzdata-2025.2-py2.py3-none-any.whl (347 kB)
Installing collected packages: pytz, tzdata, six, PyYAML, pyparsing, pillow, packaging, numpy, kiwisolver, fonttools, cycycler, python-dateutil, contourpy, pandas, matplotlib
```

Initial Project Structure Ready:

Organized the project into separate files:

- parsers.py → parsing Apache & SSH logs
- detectors.py → brute-force / DoS / scanning detection logic
- visualize.py → charts for traffic & IPs
- report.py → CSV/JSON exports
- analyzer.py → main script to integrate everything

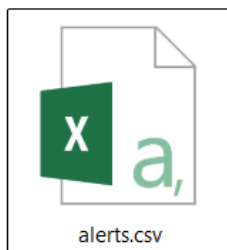
 analyzer.py	23-08-2025 22:34	Python Source File	4 KB
 config.yaml	23-08-2025 22:54	Yaml Source File	1 KB
 detectors.py	23-08-2025 22:34	Python Source File	4 KB
 parsers.py	23-08-2025 22:34	Python Source File	3 KB
 README.md	23-08-2025 22:34	Markdown Source ...	2 KB
 report.py	23-08-2025 22:34	Python Source File	1 KB
 requirements.txt	23-08-2025 22:34	Text Document	1 KB
 visualize.py	23-08-2025 22:34	Python Source File	2 KB

Test Run with Sample Logs:

Successfully ran the analyzer on **sample Apache & SSH logs**.

Output generated: alerts.csv, alerts.json, and traffic charts.

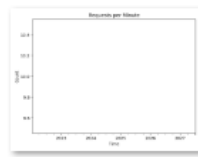
```
C:\log-file-analyzer>python analyzer.py --apache data/sample_access.log --ssh data/sample_auth.log --blacklist data/blacklist.csv --out outputs
C:\Users\hari6\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\LocalCache\local-packages\Python312\site-packages\pandas\plotting\_matplotlib\core.py:1561: UserWarning: Attempting to set identical low and high xlimits makes transformation singular; automatically expanding.
  ax.set_xlim(left, right)
Parsed events: 17
Alerts: 1
Outputs written to: outputs
C:\log-file-analyzer>
```



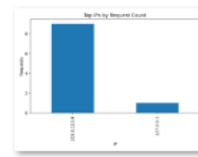
alerts.csv



alerts.json



requests_over_time.png



traffic_by_ip.png

Current Status:

I have completed the **environment setup** and built the initial project structure in Python.

The tool is able to **parse sample Apache and SSH logs** and successfully **detects brute-force and scanning patterns**. It generates **alerts in CSV/JSON** format and visualizes traffic using matplotlib charts.

Currently testing with sample logs, and the **next step** is to run the tool on real log files, adjust detection thresholds, and prepare the final 2-page project report.